

Parallel Protein Sequence Analysis and Functional Classification Using Machine Learning

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INTRODUCTION

- Proteins are essential biological molecules responsible for various cellular functions.
- The function of a protein is determined by its amino acid sequence.
- With the rapid growth of protein databases, manually annotating protein functions has become difficult.
- Machine Learning can learn patterns from protein sequences and predict:
 1. Protein family
 2. Enzyme class
 3. Gene Ontology (GO) terms
 4. Biological functions

PROBLEM STATEMENT

Given:

- A protein amino acid sequence.

Example: P12345 | MKTIIALSYIFCLVADYKDDDDDK...

Goal:

- Predict the functional category of the protein.

Example Output:

Protein Function: Oxidoreductase activity

Protein Family: Enzyme Family X

PROBLEM STATEMENT

Scale Problem: Protein databases have:

- Pfam: 46,872 protein sequences
- Manual annotation: Impossible at this scale

Computational Problem:

- Feature extraction per sequence: ~10-50 ms
- $46,872 \text{ sequences} \times 10\text{ms} = 468 \text{ seconds}$ for single feature type
- Multiple feature types + ML training = hours of computation
- Need for parallel processing

Functional Prediction Problem:

- Given: Amino acid sequence (ex- MKVLWAALLVTFLAG...)
 - Predict: Protein family/function (ex- "7-transmembrane receptor")
 - Challenge: Learn patterns from 62 protein families with imbalanced classes
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LITERATURE REVIEW

Paper Name	Author	Year	Methodology Used	Limitations
Implementing and Accelerating HMMER3 Protein Sequence Search on CUDA	Lin Cheng	2014	HMMER3 for classification, Texture memory (for speedup)	Random memory access - when the GPU has to jump around too much to find data, it slows down drastically
GPU-Accelerated Primal Learning for Large-Scale Classification	Halloran & Rocke	2020	Optimized the TRON algorithm for SVM and Logistic Regression on proteomics data using CUDA.	The algorithm is "sequentially dependent." This means Step B cannot start until Step A is finished and the data is sent back to the CPU. - GPU kept idle for too long.
GPU Technology for Accelerating Local Complexity Analysis of Proteins	Papadopoulos et al.	2013	Developed GPU_CAST for masking low-complexity regions.	Moving data from the main computer memory to the graphics card took up 30% of the total time on average. GPU wasted if the sequence doesnt have low-complexity regions.
High-Performance Bioinformatics: Accelerating Evolutionary Computation	Abi Cit	2024	Utilized V100/RTX 3090 GPUs to accelerate genetic algorithms for sequence alignment.	Does not address multi-modal feature engineering; limited by the complexity of adapting non-parallel algorithms to GPU kernels. High cost of the specialized hardware (RTX 3090/V100).

OBJECTIVES

- Preprocess and analyze protein sequences.
 - Extract biologically meaningful features.
 - Classify protein families using machine learning.
 - Implement feature extraction using:
 1. Sequential CPU
 2. OpenMP CPU
 3. CUDA GPU
 - Compare execution time, speedup, efficiency, and scalability.
 - Compare ML models based on classification performance.
-

DATASET

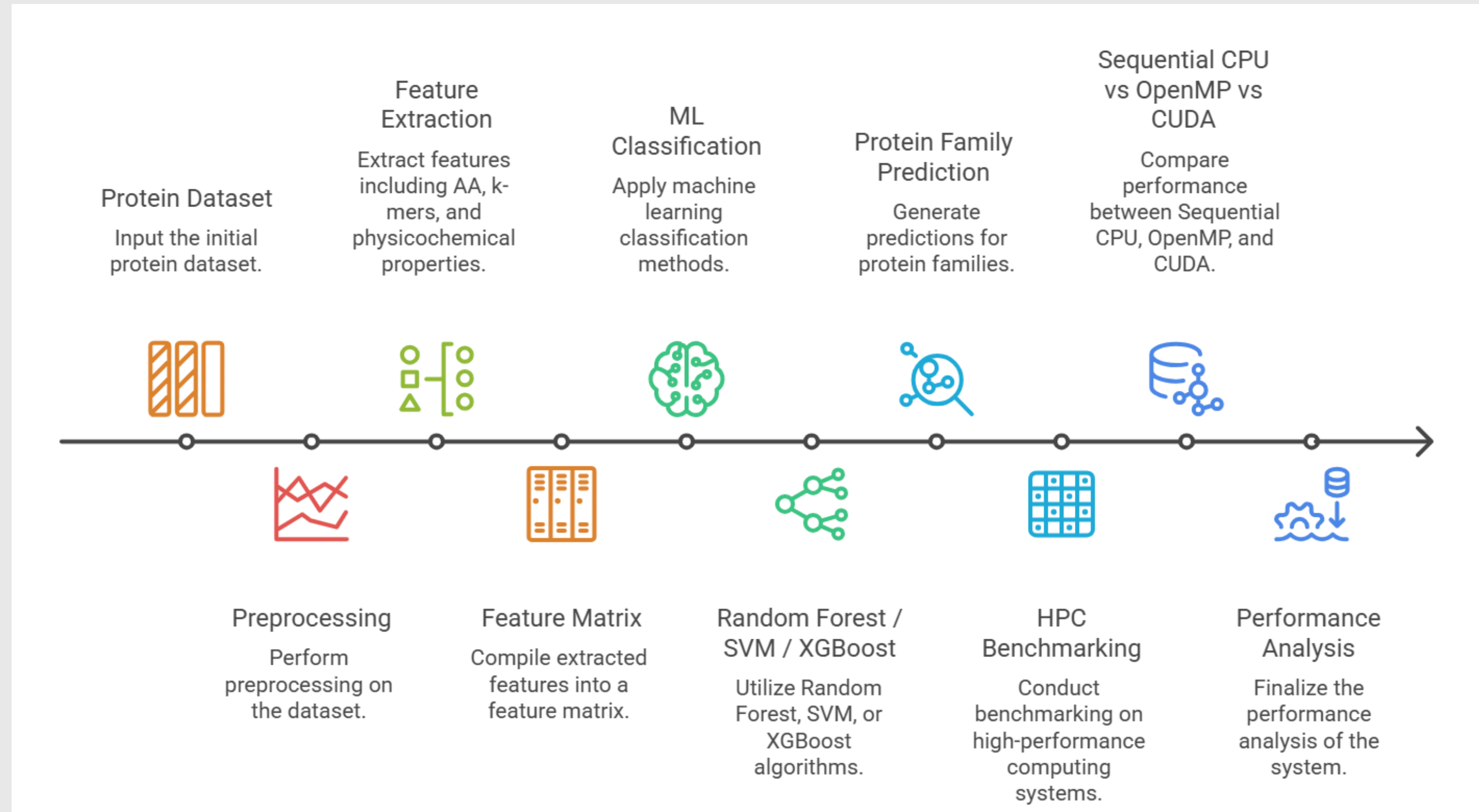
Pfam Protein Family Database

Dataset Statistics:

- Total Sequences: 46,872
- Protein Families: 62
- Sequence Length: 20 – 3,000+ residues
- Format: CSV
(structureid, classification, sequence)
- Min Sequence Length: 20
- Max Sequence Length: 3,047
- Mean Sequence Length: 287.4
- Class Imbalance Ratio: 1:100+

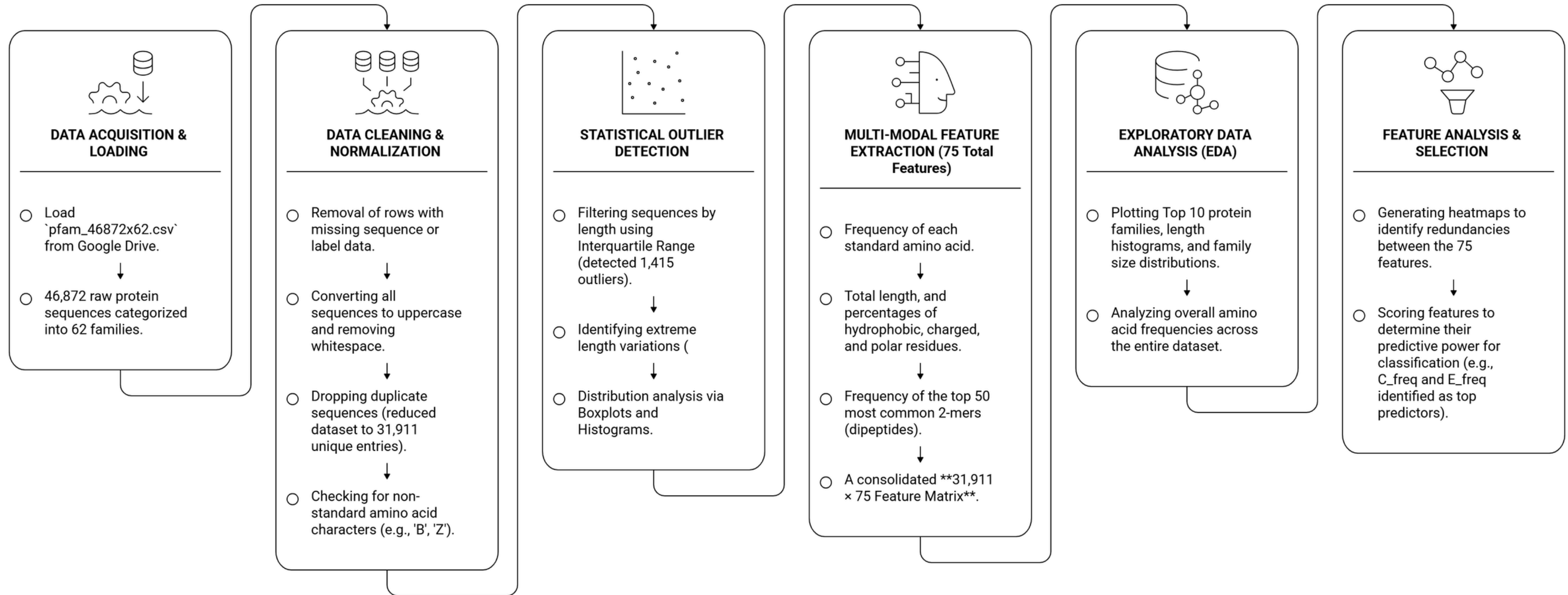
	A	B	C	D	E	F
1	structureid	classification	sequence			
2	1A9V	ALLERGEN	SQVDVKDCANHEIKKVLVPGCHGSEPCIIHRGKI			
3	1AHK	ALLERGEN	DQVDVKDCANNEIKKVMVDGCHGSDPCIIHRG			
4	1AHM	ALLERGEN	DQVDVKDCANNEIKKVMVDGCHGSDPCIIHRG			
5	1BJ7	ALLERGEN	QETPAEIDPSKIPGEWRIIYAAADNKDKIVEGGP			
6	1BV1	ALLERGEN	GVFNYETETTSVIPAARLFKAFILDGDNLFPKVA			
7	1.00E+09	ALLERGEN	GVFTYESEFTSEIPPPRLFKAFVLDADNLVPKIAP			
8	1EW3	ALLERGEN	VAIRNFDISKISGEWYSIFLASDVKEKIEENGSMF			
9	1FM4	ALLERGEN	GVFNYETEATSVIPAARMFKAFILDGDKLVPKVA			
10	1G5U	ALLERGEN	MSWQTYVDDHLMCDIDGHRLTAAAIIGHDGSY			
11	1H2O	ALLERGEN	GVFTYESEFTSEIPPPRLFKAFVLDADNLVPKIAP			
12	1ICX	ALLERGEN	GIFAFENEQSSTVAPAKLYKALTKDSDEIVPKVIE			
13	1IFV	ALLERGEN	GVFAFEDEHPSAVAQAKLTKDSDDIIPKVIE			

WORKFLOW



PREPROCESSING

Data Preprocessing & Feature Engineering



DATASET CLEANING

--- Cleaning Steps ---

After removing missing: 46872

Sequences normalized (uppercase)

After removing duplicate sequences: 31911

Invalid sequences found: 3

Examples:

MEQQPLTLTAATTRAQELRKQLNQYSHEYVVDQPSVEDY... has {'Z'}

TTPLVHVASVEKGRSYEDFQKVYNALALKLREDDEYDNYI... has {'B'}

ANSVCFTFTDFESGQQDLIFQGDASVGSNKALQLTKVDSK... has {'B'}

✓ Keeping (< 10% of data)

Final cleaned: 31911 sequences

Removed: 14961 sequences

OUTLIER DETECTION

1. IQR Method

- Finds values that are unusually far from the normal range.
- Uses the middle 50% of the data to set the normal range.
- Detected around 1415 outliers (4.43 %)

2. Z-Score Method

- Checks how far a value is from the average.
- Values more than 3 standard deviations away are treated as extreme outliers.
- Detected around 515 extreme outliers (1.61 %)

Visualization:

- Boxplot was used to clearly identify and visualize the outliers.

```
=====
OUTLIER DETECTION (Sequence Length)
=====

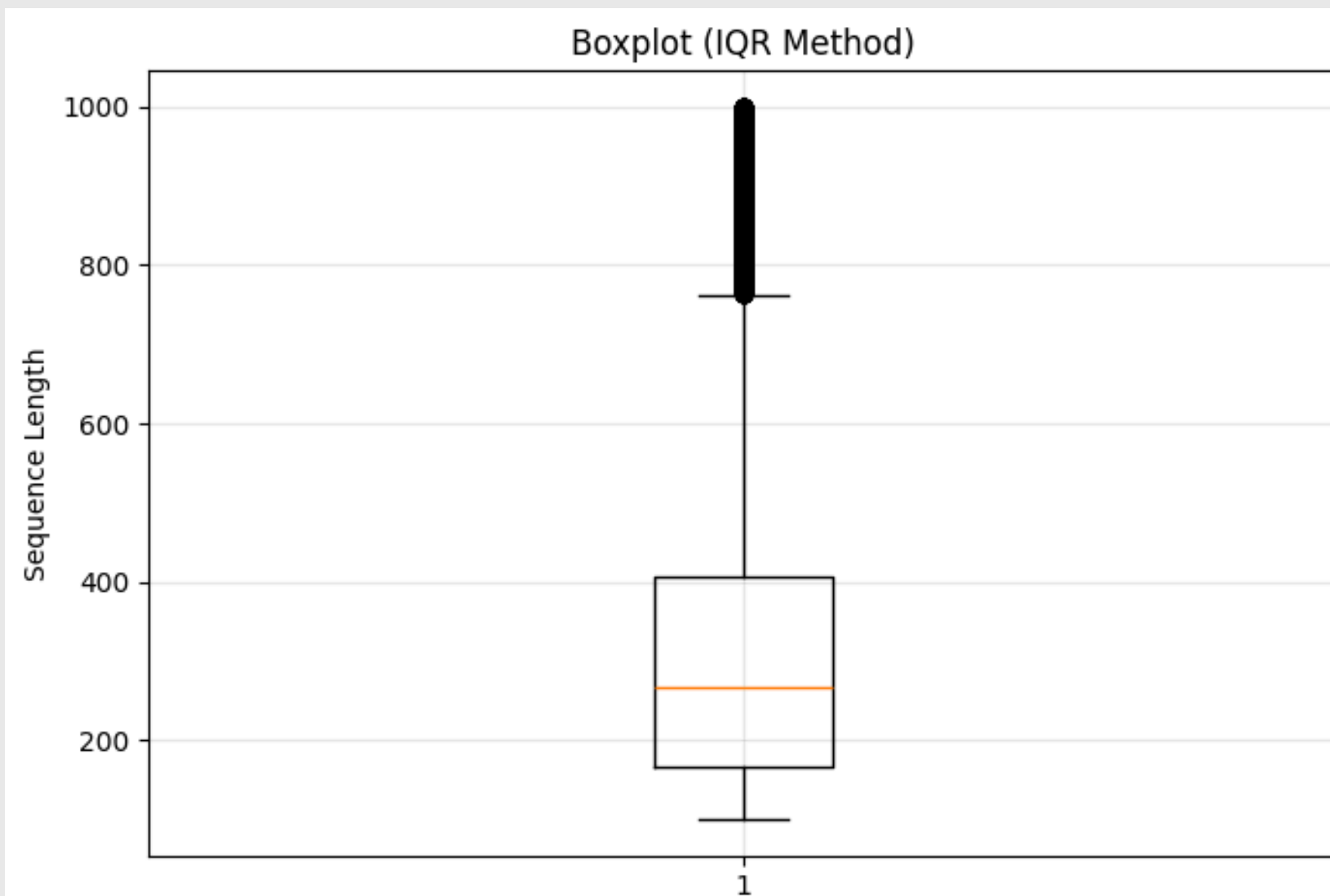
IQR Method:
  Q1=168, Q3=406, IQR=238
  Bounds: [-189, 763]
  Outliers: 1415 (4.43%)

Z-Score Method (|Z| > 3):
  Outliers: 515 (1.61%)

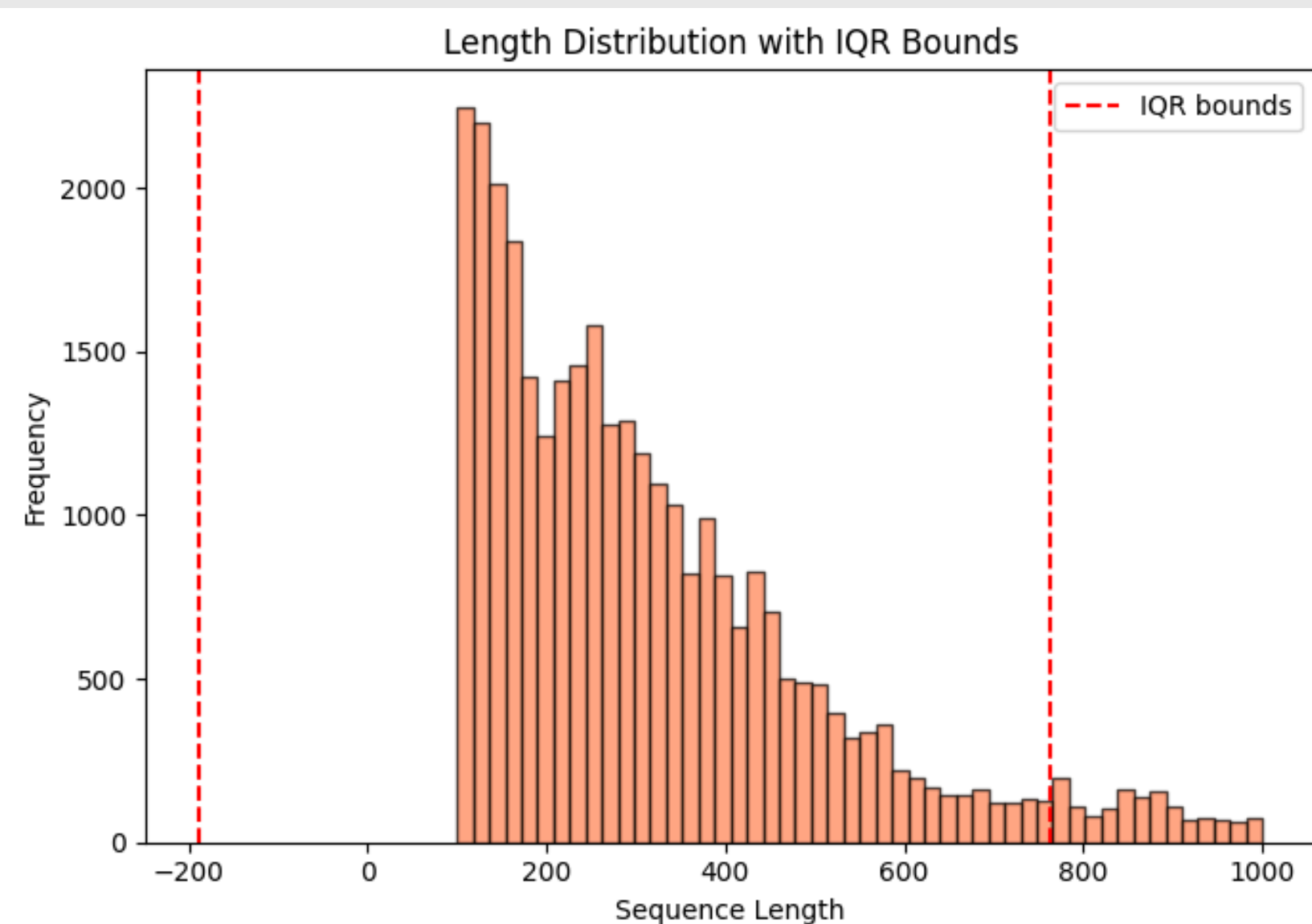
Combined: 1415 outliers detected
```

OUTLIER DETECTION

Box Plot

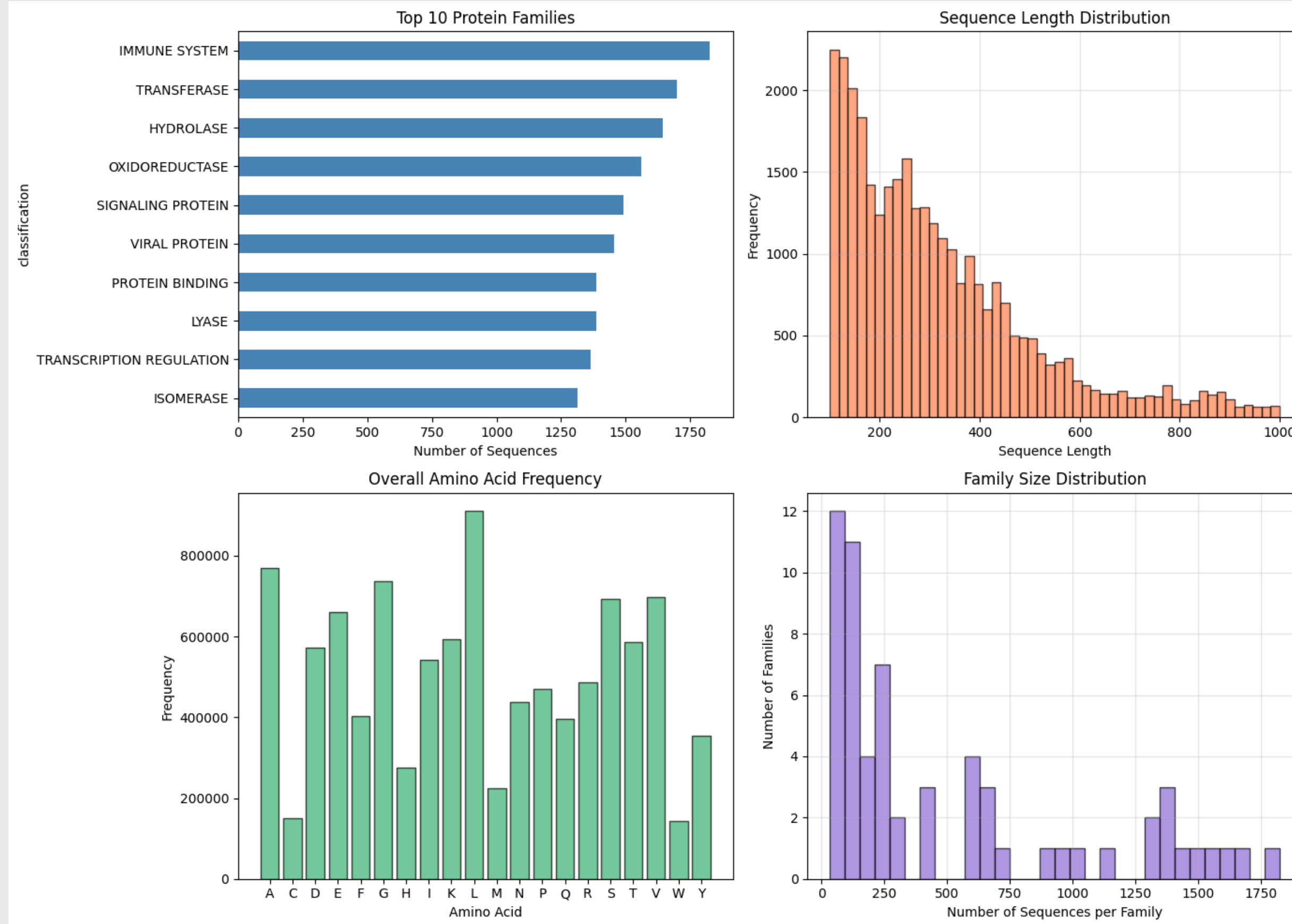


Histogram



30496 Sequences remain after removing outliers

EDA



FEATURE EXTRACTION – Multi-Modal Feature Engineering

10

3 Feature Categories:

1. Amino Acid Composition – 20 features

- Counts how much of each amino acid is present.
- Example: 15% Alanine, 22% Leucine, etc.
- Gives the protein's basic composition/fingerprint.

2. Sequence Properties – 5 features

- Sequence length
- Hydrophobicity
- Charge
- Polarity
- Aromaticity
- Helps describe the physical and chemical nature of the protein.

3. k-mer Features – 50 features

- Looks at small groups of 2 amino acids (dipeptides) – AL, LE, ER.
- Measures how often these patterns appear.
- Helps capture common sequence patterns/motifs.
- Only the top 50 are used.

```
1. Amino Acid Composition (20 features)
  ✓ 20 features
```

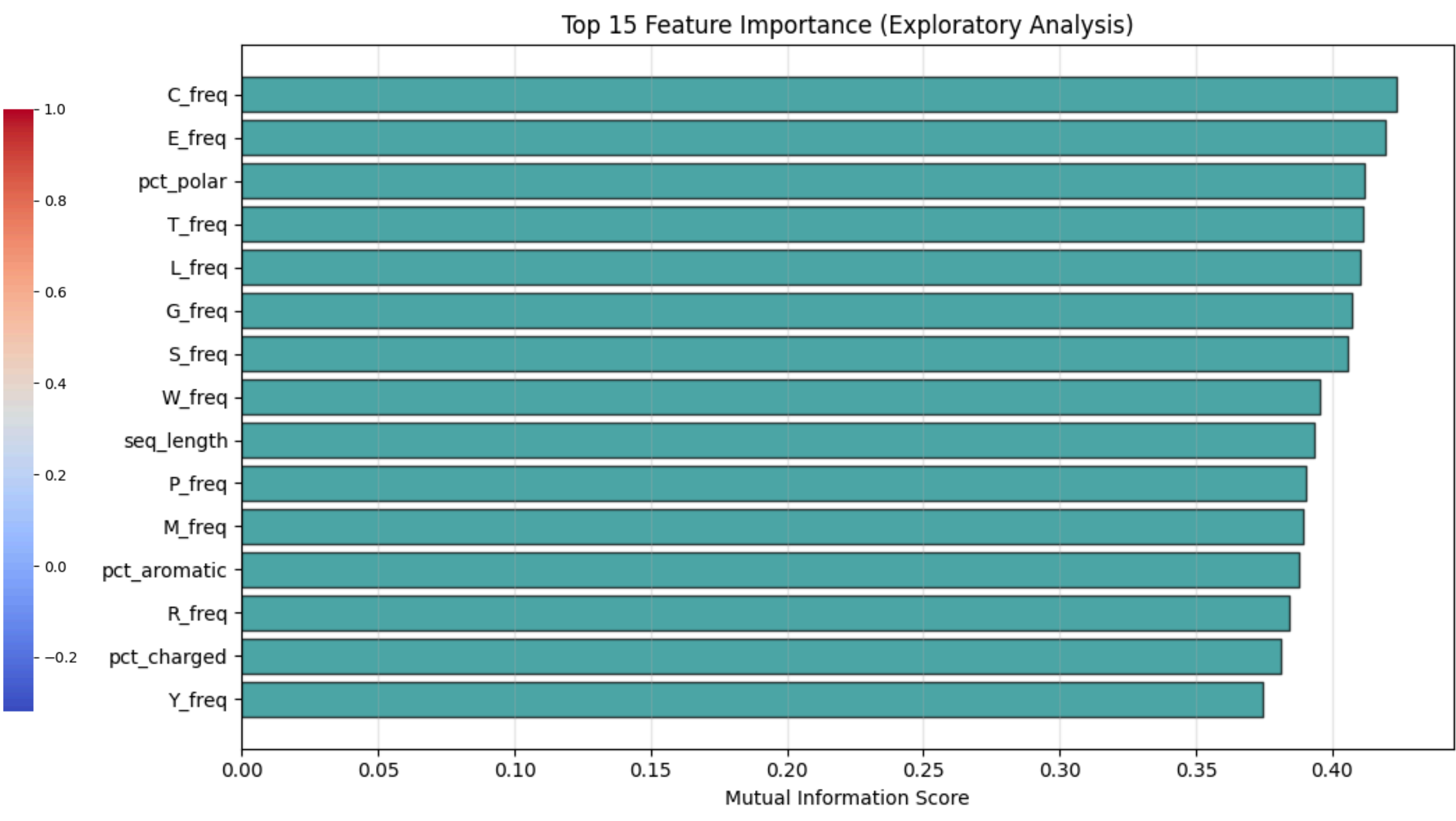
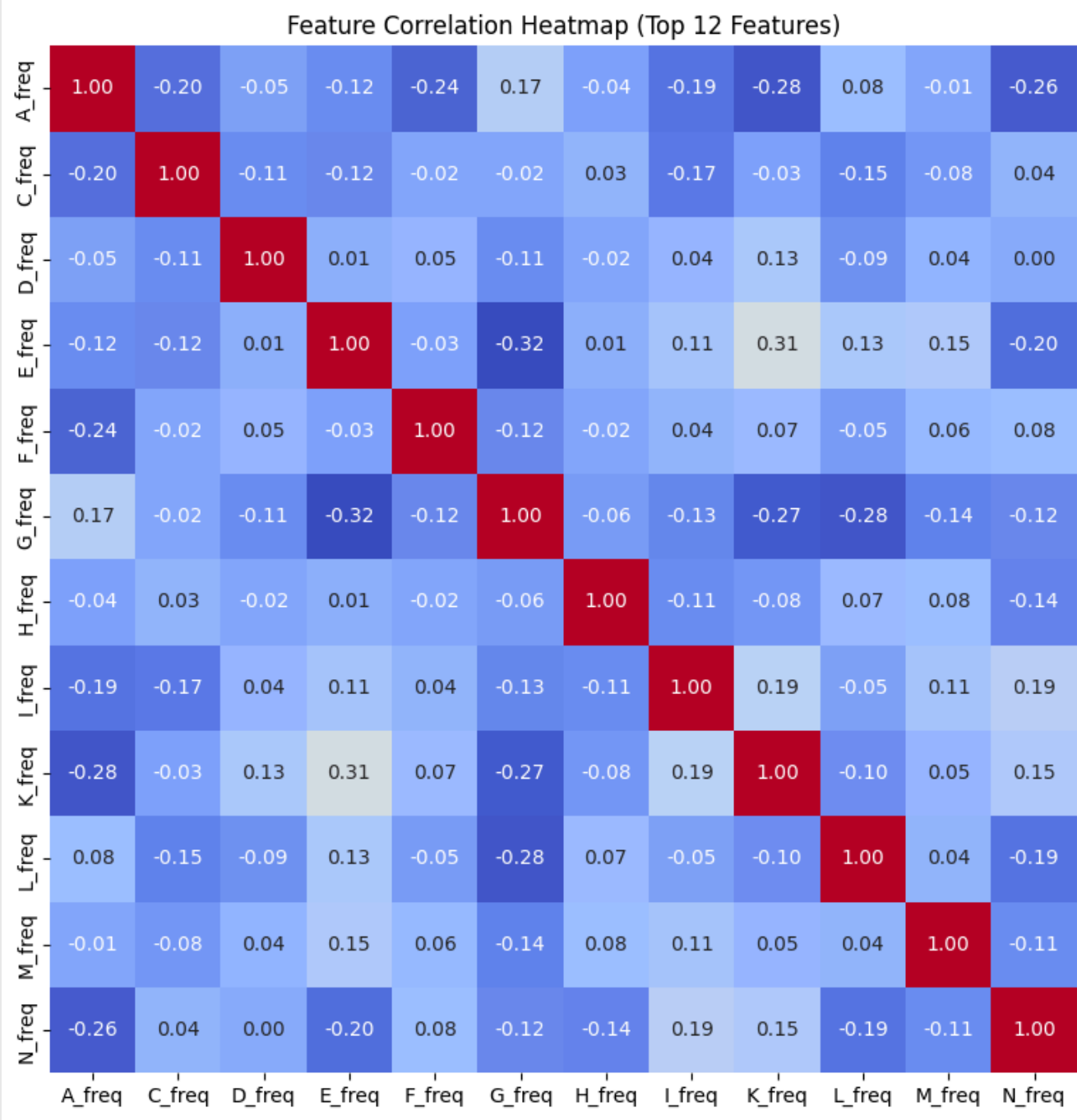
```
2. Sequence Features (5 features)
  ✓ 5 features
```

```
3. k-mer Features (default k=2)
   Extracting 2-mers...
   Using top 50 2-mers out of 461 unique
   ✓ 50 features
```

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✓ TOTAL FEATURES: 75
```

```
✓ Feature matrix: 31911 x 75
```

FEATURE ANALYSIS



No strong correlation across most features

ML Classification

Planned Classifiers:

- Random Forest

Handles class imbalance well

Feature importance built-in

Expected accuracy: 80-85%

- Support Vector Machine (SVM), Non-linear kernels (RBF)

Good for high-dimensional data

Expected accuracy: 75-80%

- XGBoost

Handles imbalanced data

Fast training

Expected accuracy: 85-90%

HPC & Parallelization

Scaling with High Performance Computing

Parallelize:

- Feature extraction.
- Similarity calculations.
- Model training.

Compare:

- Sequential CPU
- OpenMP CPU - Expected Speedup: 8-16× on 8-16 core CPU
- CUDA GPU - Expected Speedup: 50-100× for dense operations

Metrics:

- Execution time
- Speedup
- Efficiency
- Scalability

Citations for Literature Review

- Cit, A. (2024). High-Performance Bioinformatics: Accelerating Evolutionary Computation with GPUs. EasyChair Preprint № 13983.
- Papadopoulos, A., et al. (2013). GPU Technology As a Platform for Accelerating Local Complexity Analysis of Protein Sequences. IEEE EMBS.
- Halloran, J. T., & Rocke, D. M. (2020). GPU-Accelerated Primal Learning for Extremely Fast Large-Scale Classification. NeurIPS.
- Cheng, L. (2014). Implementing and Accelerating HMMER3 Protein Sequence Search on CUDA-Enabled GPU. Concordia University Thesis.

THANK YOU